CausalCache: Restoration-Guided Event Selection for Long-Horizon GUI Memory

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Abstract

Long-horizon GUI agents must decide both how to use and which historical visual evidence to expose under a limited multimodal context budget. Existing memory controllers commonly rely on recency, similarity, attention, or salience, while ordinary policy adaptation can change behavior even when no history is present. We introduce CausalCache, which couples a history-gated policy adapter with a set-conditioned memory selector. The adapter modifies only the final eight language-model key/value projections and activates its low-rank residual only on restored history-image tokens; with no history, it structurally bypasses the adapter and exactly recovers the frozen policy. The selector estimates the conditional marginal utility of each remaining event given the already selected set and may stop before exhausting the budget. All adapters and selectors are trained and selected exclusively on GUI-Odyssey, then evaluated zero-shot on AndroidWorld and OSWorld across fixed visual-memory budgets. We separately measure end-to-end task performance and selector-specific set utility, oracle recovery, regret, stopping behavior, and latency. **[Pending: Closed-loop and selector-mechanism results are being populated under the frozen protocol.]**

Introduction

A GUI trajectory can contain screenshots, actions, text arguments, coordinates, and state changes accumulated over dozens of decisions. Passing every event to a multimodal policy is expensive and can amplify context rot, while replacing all screenshots with summaries discards exact visual evidence. A memory controller must therefore decide which historical events deserve high-fidelity exposure at each decision.

Prior controllers select memory using recency, retrieval similarity, attention, learned salience, or task-state heuristics. These signals need not agree with the frozen action policy's actual dependence on an event. Two memories can even have similar successor-action negative log-likelihood (NLL) while supporting different closed-loop outcomes. This motivates a narrower question: *how much does adding one event's archived post-action image to an otherwise unchanged strong low-fidelity history restore the frozen policy's stable full-history action behavior?*

CausalCacheanswers this question with two aligned components. A history-gated key/value adapter teaches the policy to use restored historical images without changing its no-history behavior. A selector then scores each candidate conditioned on the current query, the already selected set, and the remaining budget. It maintains a raw event archive, a cheap summary/index, and a limited policy-visible high-fidelity context. All events retain byte-identical low-fidelity summaries in every coalition; selecting an event adds only its post-action image. The selector's conditional targets are computed by rerunning the complete selected set, so redundancy and complementarity are not replaced by sums of singleton gains. Independent marginal scoring remains an explicit baseline, while exact subset search remains an offline oracle.

Our evaluation is organized around four falsifiable questions:

1. Does history-gated adaptation outperform ordinary policy adaptation when every method receives the same recent memory?
2. Does set-conditioned selection outperform recent and marginal selection under a fixed history-gated policy?
3. Do these gains persist across memory budgets and transfer zero-shot to both AndroidWorld and OSWorld?
4. How much exact or approximate oracle utility does the deployed selector recover, at what regret, STOP rate, and latency?

Related Work

GUI memory. MementoGUI learns multimodal write, compression, and retrieval operators for a frozen GUI action model (Zeng et al. 2026). AndroTMem retrieves anchored intermediate states connected to the current subgoal (Shi et al. 2026). These systems establish the importance of learned controllers and dependency-critical events. CausalCacheinstead derives selector supervision from the marginal behavior change induced by restoring the high-fidelity version of the same event.

Interventional memory selection. CMI compares agent behavior under memory interventions (Srivastava 2026), while ATMem estimates how active memory states affect

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GUI task completion (Liu et al. 2026). Our scope is intentionally narrower: we hold the action policy fixed, intervene on low- versus high-fidelity GUI evidence, derive selector supervision from coalition-conditioned behavior changes, and validate the dense decision-time surrogate through executable closed-loop outcomes.

Coalition attribution and subset selection. Permutation marginal contributions originate in cooperative game theory (Shapley 1953). We instead use a frozen, explicitly defined projection of empty- and singleton-conditioned marginals for the deployed independent selector. Full coalition tables, interaction statistics, and exact subset search serve as offline supervision analyses and upper bounds.

Problem Formulation

Let x_t be the current observation at decision t, and let its history be

$$H_t = \{e_1, \dots, e_{t-1}\}, \qquad e_j = (x_j, a_j, x_{j+1}, \delta_j), \quad (1)$$

where δ_j is an optional deterministic UI delta. Every event has an archived high-fidelity visual increment and a deterministic low-fidelity record $\tilde{e}_j$. We exclude a redundant candidate when its post-state is already the current observation:

$$C_t = \{j < t : \mathrm{post}(e_j) \not\equiv x_t\}. \quad (2)$$

A query-time selector chooses which candidate images to expose:

$$M_t \subseteq C_t, \qquad \sum_{j \in M_t} c_j \leq B, \quad (3)$$

where c_j is the visual-token cost and B is the policy-visible context budget. Every event, including events outside C_t, contributes the same low-fidelity record in every memory. Selecting $j \in C_t$ adds its post-action image and nothing else. Raw events are retained in an external archive, so an event can be retrieved again at a later query; B is not a persistent storage or eviction budget.

The base policy π_0 remains frozen inside the history-gated system; only the added low-rank adapter and selector are trained. Our primary goal is to improve executable task success at matched policy-visible budgets without attributing gains to extra visual tokens. We evaluate $B \in \{0, 1, 2, 4, 8\}$ separately.

CausalCache

History-Gated KV Adapter

For each of the final eight language-model layers, CausalCacheadds rank-8, alpha-16 residuals only to the key and value projections:

$$\begin{aligned} K_l &= W_{K,l} h + M_{\mathrm{hist}} \Delta W_{K,l} h, \\ V_l &= W_{V,l} h + M_{\mathrm{hist}} \Delta W_{V,l} h. \end{aligned} \quad (4)$$

The token-role mask M_{hist} is true only for restored historical image tokens and excludes the current observation, text, and generated action tokens. When $B = 0$, the mask is empty and the residual path is bypassed rather than multiplied by zero after evaluation. This makes the history-gated policy structurally identical to frozen GUI-Owl at $B = 0$. The architecture-matched ungated control uses the same layers, projections, rank, alpha, data, and loss, but applies the residual to every token.

Two-Level Event Representation

The deterministic low-fidelity record contains

```
step_id, action_type, action_argument,
foreground_app,
screen_text_added, screen_text_removed,
screen_change,
executor_result.
```

The record is present for every event in every coalition and its serialization is byte-identical between low- and high-fidelity conditions. Accessibility content, or a pinned OCR fallback, supplies the normalized screen-text delta. For a selected event, the policy-visible high-fidelity increment is exactly one post-action image: no before image, additional action text, or raw-argument text is added. The raw archive may retain both screenshots and execution metadata, but those fields are not part of the restoration intervention. We report the actual policy-visible text tokens rather than assuming exactly fixed text cost. The frozen gate input is the 200-dimensional vector $[q_{64}, h_{64}, q_{64} \odot h_{64}, g_8]$. Here q_{64} hashes the instruction and current OCR tokens, h_{64} hashes the event summary and candidate OCR tokens, and g_8 contains bounded age, action, OCR, change, and result geometry. Policy-vision features and the requested budget are not gate inputs. For live AndroidWorld histories, an oldest-to-newest adapter rebuilds the same record from the canonical executed action, pinned OCR records, screen-text delta, and 256×256 RGB difference. It assigns age as history length minus one minus chronological index, excludes only the newest current-equivalent post-state, and scores every older event. Fields unavailable in the offline source (foreground app and executor result) remain the same fixed `unknown` values rather than introducing a train–test feature shift. The action policy receives mixed-fidelity inputs through a normal forward pass; we do not splice cached keys and values across incompatible contexts.

Stable Full-History Self-Behavior Reference

Let $X_t(S)$ denote the shared low-fidelity history with post-action images added only for $S \subseteq C_t$. The frozen-policy reference is

$$q_t = \pi_0(\cdot \mid g, x_t, X_t(C_t)). \quad (5)$$

Its deterministically decoded output must be one exact native MobileUse tool call and is used as the canonical action a_t^*. The substrate requires contract-parseable output, finite teacher-forced logits, and identical canonical actions across two deterministic forwards. Expert agreement does not define or filter the reference. Once the untouched confirm split is opened, parse or stability failures count against the fixed-denominator gate rather than being removed. We use the checkpoint's official tool-template path rather than reproducing its schema in a hand-written prompt. For teacher forcing, the processor's native assistant prefix is followed directly by the deterministic MobileUse tool call; no generated action description or fixed Action carrier is inserted. Text arguments must already use Unicode NFKC normalization.

Executable Behavioral Distance

For a coalition S, all events retain their low-fidelity records, and only events in S add images. We measure pathwise divergence along the canonical action rather than claiming a full sequence-action KL:

$$D_t(S) = \frac{1}{L} \sum_{l=1}^{L} \mathrm{KL}\big(q_{t,l}(\cdot \mid a^*_{t,<l}) \,\|\, q^S_{t,l}(\cdot \mid a^*_{t,<l})\big)\,. \quad (6)$$

The KL is teacher-forced over the full vocabulary from the opening <tool_call> through the model-emitted closing tag. Standard model EOS tokens that are suppressed during generation receive the same finite minimum logit mask in both reference and candidate distributions before normalization. Action-type match, canonical coordinate-bin match, normalized coordinate distance, and text-argument match are reported as diagnostics, not mixed into the primary restoration distance.

Restoration Targets for Independent Selection

For event j, its conditional marginal under an already restored coalition S is

$$\Delta_{j,t}(S) = D_t(S) - D_t(S \cup \{j\}). \quad (7)$$

These marginals retain redundancy and complementarity because the frozen policy is rerun on the complete mixed-fidelity coalition. Our deployed student is deliberately simpler: it predicts one query-dependent value per event. The frozen independent target averages the empty-coalition and singleton-conditioned marginals,

$$G^{\mathrm{ind}}_{j,t} = \frac{1}{2}\left[\Delta_{j,t}(\varnothing) + \frac{1}{|C_t|-1} \sum_{i \in C_t \setminus \{j\}} \Delta_{j,t}(\{i\})\right]. \quad (8)$$

This is an event-level projection of an interacting set utility, not a claim that events are independent. The regression target is $G^{\mathrm{ind}}_{j,t} / \max(D_t(\varnothing), 10^{-12})$. States at or below the denominator epsilon are excluded only from normalized regression and means; their raw target, sign, ranking, and raw evaluation remain present. We measure the projection's gap to exact subset search and retain this marginal selector as the independent baseline. The proposed selector instead predicts $\Delta_{j,t}(S)$ from the current decision query, the candidate's HGKV readout, the already selected set, and the remaining budget. Conditional labels are rendered along frozen shortlist and greedy/beam paths. Every selected set used for evaluation is rerun through the history-gated policy; singleton gains are never summed and reported as if they were the complete set utility.

Query-Time Selectors

The distilled selector predicts

$$s_{j,t} = f_\theta(q_{64,t}, h_{64,j}, q_{64,t} \odot h_{64,j}, g_{8,j,t}) \quad (9)$$

Algorithm 1 Offline Restoration Labels

Require: Stable-reference state $(g, x_t, H_t, C_t, a^*_t)$

Include the shared low-fidelity record for every event
Run $X_t(C_t)$ once to cache full-distinct-history behavior q_t
for each required coalition S **do**
  Run the frozen policy on $X_t(S)$ under teacher forcing
  Record $D_t(S)$
end for
Compute $G^{\mathrm{ind}}_{j,t}$ and the exact budget-B subset oracle

with normalized regression and raw-target ranking supervision. At inference, the selector solves a cost-constrained positive-value selection problem:

$$M_t = \arg \max_{M : \sum_{j \in M} c_j \le B} \sum_{j \in M} \max(s_{j,t}, 0). \quad (10)$$

For equal-cost event blocks this reduces to selecting up to, rather than exactly, the top $b = \lfloor B/c \rfloor$ events whose scores are strictly positive. The threshold is frozen at zero before confirm access; it is not tuned on confirm or closed-loop outcomes. This permits a null decision when all candidate restorations are predicted to be harmful or useless.

The set-conditioned selector starts from $S = \varnothing$ and repeatedly predicts the conditional marginal of every remaining candidate. At each step it compares the best candidate with a learned STOP score, stopping when STOP wins or when $|S| = B$. A lightweight selected-set attention block, candidate–query attention block, marginal head, and STOP head share the same frozen HGKV readouts. The marginal selector above uses the same first-stage singleton scorer but does not condition later choices on S.

Experimental Protocol

Benchmarks and Policies

AndroidWorld (Rawles et al. 2025) and OSWorld (Xie et al. 2024) are the two closed-loop benchmarks. GUIOdyssey (Lu et al. 2025) supplies all adapter training, selector training, and model selection data. No AndroidWorld or OSWorld episode is used to choose an architecture, checkpoint, threshold, or reported budget. The evaluation is therefore zero-shot on both target benchmarks.

The base policy is GUI-Owl-1.5-8B-Instruct (Xu et al. 2026) at model revision `06d5faec`, using bfloat16 and deterministic decoding through its official tool template. All comparisons use the same low-fidelity event summaries, high-fidelity image preprocessing, parser/executor, and visual-memory accounting. A budget B denotes at most B restored historical images in addition to the current observation.

AndroidWorld uses two fixed instances for each of all 116 task templates. Its primary metric is task-template macro success rate: we first average the two instance outcomes within each template and then average the 116 template means. Infrastructure failures remain zero in the fixed intention-to-treat denominator. OSWorld uses the full fixed evaluation roster and reports mean normalized task score. For both benchmarks, "Avg. $B > 0$" is the arithmetic mean of the separately aggregated $B \in \{1, 2, 4, 8\}$ scores.

At $B = 0$, the frozen row is evaluated once and the three HGKV rows are structurally identical to it because the history-gated residual is bypassed. Full-layer LoRA and top-8 ungated KV LoRA require separate $B = 0$ evaluations because both can alter the current-observation policy even without history.

Comparisons

The end-to-end table has six rows rather than a policy–selector Cartesian product. Frozen GUI-Owl, full-layer LoRA, top-8 ungated KV LoRA, and HGKV are first compared under the same Recent selector. Holding HGKV fixed, Recent, marginal, and set-conditioned selection then isolate the value of modeling interactions among selected events. Random, similarity, full-history, oracle, and architecture/training sweeps are diagnostics or appendix ablations.

Success–Memory Frontier

We report the full closed-loop curve at $B \in \{0, 1, 2, 4, 8\}$ rather than a single favored budget. Every unavailable cell remains an explicit TBD until its frozen-roster aggregate is complete.

Legacy Matched-NLL Mechanism Test

This analysis is not part of the two primary tables. The following text records the legacy frozen protocol; applying it to HGKV would require a separately preregistered version with a new source and untouched holdout. After each authorized partition's immutable primary closed-loop report, we use the policy-decision states from both origin episodes without deduplicating across origins. At each state, both memories are evaluated against a canonical action generated twice from their visual union, which contains at most four historical images plus the current observation. This pair-union reference makes the symmetric comparison executable under the five-image policy envelope. We compute mean teacher-forced action-token NLL and raw restoration mass $R_m = D_t(\varnothing) - D_t(M_m)$ for each memory. Aggregation is equal-state within each origin episode, equal-half over the two origin episodes, equal-instance within a template, then equal-template. A template is mechanism-eligible only when every preregistered instance pair and every one of its decision states passes the frozen repeat, finite-metric, selection, and budget audits.

The primary matched caliper is fixed in advance at 0.05 nats per token; 0.02 and 0.10 are sensitivity analyses. Within NLL-matched templates, the directional statistic is $\mathrm{sign}_{10^{-12}}(R_{\mathrm{ind}} - R_{\mathrm{recent}})(Y_{\mathrm{ind}} - Y_{\mathrm{recent}})$, with restoration ties contributing zero. The train-60 analysis is development-only. We make a primary mechanism claim only on sealed-test-75, only if at least 15 matched template clusters remain and the frozen 90% template-bootstrap lower bound is positive. Reference or metric failures make a template ineligible only for this mechanism analysis; they remain in the closed-loop intention-to-treat denominator. We never choose a caliper from the observed success or NLL distribution. This is a post-treatment mechanism association on controller-conditioned state distributions, not a causal mediation effect.

Attribution Cost and Stability

We vary coalition samples K and report attribution standard error, Spearman rank correlation to a high-K estimate, top-budget Jaccard overlap, oracle success, total offline labeling compute, and online selector latency. The per-permutation telescoping identity is used only as an implementation sanity check. Interaction analyses use the already frozen development studies rather than a statistic invented after confirm access. The confirm experiment uses the exact 16-coalition table for its four candidates. Its primary statistics aggregate raw utility over the fixed 20-state denominator; $D_t(\varnothing) > 10^{-12}$ is used only to count memory-sensitive states. No small-denominator state is removed or promoted to a separate primary average.

Prior independent-selector diagnostic. The earlier confirm report covered all 20 states and retained a substantial exact restoration-oracle signal, but the learned independent selector reached only 0.8213 of exact raw utility and did not beat the strongest heuristic. A read-only decomposition over the already consumed complete coalition tables gave raw utility sums $0.8564/0.7833/0.7653/0.7034$ for exact, OCR/RGB, oracle-independent projection, and learned independent selection, respectively. The oracle-independent minus OCR/RGB raw mean was -0.00090, whereas oracle-independent minus learned selection had a 90% paired interval $[0.00048, 0.00653]$. Hence both projection and distillation contributed error in that version; the latter was not the only blocker. The consumed confirm set is not reused for the current HGKV selector or either zero-shot closed-loop benchmark.

Limitations and Claim Boundary

Restoration value is a dense decision-time surrogate. It is not a direct long-horizon causal effect, and its relevance to long-horizon control must be established through closed-loop success and the matched-NLL analysis. The causal language in this paper refers only to a controlled low-/high-fidelity intervention on a frozen policy's behavior. We do not identify environment-level structural causality and do not learn a world model. The method should be weakened or abandoned if restoration fails to beat recency, similarity, and attention; if its matched-NLL association disappears; if the distilled gate cannot approach the oracle; or if gains come from extra tokens. A parseable and repeat-stable self-behavior reference can still be consistently wrong with respect to an expert or the environment. Restoration therefore measures fidelity to a frozen policy, not action correctness. Expert alignment and executable closed-loop success remain separate diagnostics and validation requirements.

Our offline experiments partition GUIOdyssey trajectories with a project-specific split: roles are assigned at the trajectory level by deterministic hashing with instruction–app group deduplication and history-length stratification, frozen

Table 1: Zero-shot closed-loop performance under different visual-memory budgets. All adapters and selectors are trained and selected exclusively on GUI-Odyssey. AndroidWorld uses two fixed instances for each of 116 templates; OSWorld uses the full fixed evaluation roster. HGKV is structurally identical to the frozen policy at $B = 0$.

Policy adaptation	Memory selector	$B = 0$	$B = 1$	$B = 2$	$B = 4$	$B = 8$	Avg. $B > 0$
Panel A: AndroidWorld — task-template macro success rate (%)							
Frozen GUI-Owl	Recent	TBD	TBD	TBD	TBD	TBD	TBD
Full-layer LoRA	Recent	TBD	TBD	TBD	TBD	TBD	TBD
Top-8 ungated KV LoRA	Recent	TBD	TBD	TBD	TBD	TBD	TBD
HGKV adapter	Recent	= Frozen	TBD	TBD	TBD	TBD	TBD
HGKV adapter	Marginal selector	= Frozen	TBD	TBD	TBD	TBD	TBD
HGKV adapter	**Set-conditioned selector (Ours)**	= Frozen	TBD	TBD	TBD	TBD	TBD
Panel B: OSWorld — mean normalized task score (%)							
Frozen GUI-Owl	Recent	TBD	TBD	TBD	TBD	TBD	TBD
Full-layer LoRA	Recent	TBD	TBD	TBD	TBD	TBD	TBD
Top-8 ungated KV LoRA	Recent	TBD	TBD	TBD	TBD	TBD	TBD
HGKV adapter	Recent	= Frozen	TBD	TBD	TBD	TBD	TBD
HGKV adapter	Marginal selector	= Frozen	TBD	TBD	TBD	TBD	TBD
HGKV adapter	**Set-conditioned selector (Ours)**	= Frozen	TBD	TBD	TBD	TBD	TBD

Table 2: Selector mechanism analysis on GUI-Odyssey development states under the fixed HGKV policy. Utility at each budget is measured by rerunning the complete selected set through HGKV; it is not a sum of singleton gains. Oracle recovery and regret use exact search for small candidate sets and a declared approximate oracle otherwise.

Selector	$B = 1$ utility	$B = 2$ utility	$B = 4$ utility	Oracle recovery	Oracle regret	STOP rate	Latency
Random	TBD	TBD	TBD	TBD	TBD	TBD	–
Recent	TBD	TBD	TBD	TBD	TBD	TBD	–
Similarity	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Marginal HGKV selector	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Set-conditioned HGKV selector	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Oracle / approximate oracle	TBD	TBD	TBD	1.000	0	–	–

before any label generation. This partition deliberately differs from the four official GUIOdyssey splits, which are two-way, are not length-stratified, and allow near-duplicate instruction–app combinations to cross train and test; a released audit maps every trajectory role onto the official lists. Because all offline metrics are internal paired comparisons over restoration labels that we generate, no official-leaderboard comparison is made or implied. We also cannot rule out that the frozen policy's pretraining corpus contains GUIOdyssey episodes; the protocol isolates selector training from selector evaluation but claims no decontamination of the policy itself. Neither caveat affects the primary closed-loop results, which compare frozen-base policy variants and memory controllers on the fixed AndroidWorld and OSWorld rosters.

Conclusion

CausalCachecombines a no-history-preserving HGKV adapter with interaction-aware visual-memory selection. Its central empirical burden is whether history gating outperforms ordinary adaptation under the same memory, and whether set-conditioned selection improves the closed-loop success–memory frontier beyond recent and marginal selection. The zero-shot AndroidWorld and OSWorld tables keep every unfinished result explicit; no empirical claim is made from a TBD cell.

Planned Ablations

The end-to-end main table remains limited to the six comparisons needed to separate policy adaptation from memory selection. The following families are reserved for the appendix and will be reported in separate metric-appropriate tables rather than expanded into a policy–selector Cartesian product.

Table 3: Appendix ablation placeholders. Each family is evaluated with the corresponding policy, selector, and data split fixed.

Ablation family	Planned variants	Status
Closed-loop selector controls	Random; similarity; full-history ceiling; oracle / approximate oracle	TBD
HGKV depth	Adapters in the final 4, 8, or 12 language-model layers	TBD
HGKV rank	Rank 4, 8, or 16 with matched target modules and gating	TBD
Set encoder	DeepSets; lightweight set attention; Set Transformer	TBD
Training objective	Frozen loss-weight variants under the same data and checkpoint budget	TBD
Robustness corruptions	Selected top-B vs. bottom-B; wrong-event replacement; summary–image mismatch; temporal reversal	TBD

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